Technology Brief 8: Supercapacitors as Batteries

As recent additions to the language of electronics, the names ***supercapacitor***, ***ultracapacitor***, and ***nanocapacitor*** suggest that they represent devices that are somehow different from or superior to traditional capacitors. Are these just fancy names attached to traditional capacitors by manufacturers, or are we talking about a really different type of capacitor?

► The three aforementioned names refer to variations on an energy storage device (**Fig. TF8-1**) known by the technical name ***electrochemical double-layer capacitor*** (***EDLC***), in which *energy storage is realized by a hybrid process that incorporates features from both the traditional electrostatic capacitor and the electrochemical voltaic battery.* ◄

For the purposes of this Technology Brief, we refer to this relatively new device as a supercapacitor. The battery is far superior to the traditional capacitor with regard to energy storage, but a capacitor can be charged and discharged much more rapidly than a battery. As a hybrid technology, the supercapacitor offers features that are intermediate between those of the battery and the traditional capacitor. The supercapacitor is now used to support a wide range of applications, from motor startups in large engines (trucks, locomotives, submarines, etc.) to flash lights in digital cameras, and its use is rapidly extending into consumer electronics (cell phones, MP3 players, laptop computers) and electric cars (**Fig. TF8-2**).

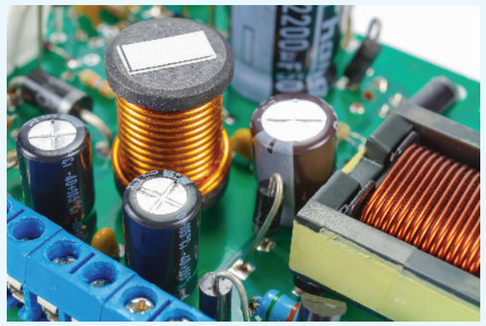


Figure TF8-1 Examples of supercapacitors. (Courtesy of Vladimir Zhupanenko/Shutterstock.)

Capacitor Energy Storage Limitations

Energy density W' is often measured in watts-hours per kg (Wh/kg), with $1\ \text{Wh} = 3.6 \times 10^3$ joules. Thus, the energy capacity of a device is normalized to its mass. For batteries, W' extends between about 30 Wh/kg for a lead-acid battery to as high as 150 Wh/kg for a lithium-ion battery. In contrast, W' rarely exceeds 0.02 Wh/kg for a traditional capacitor. Let us examine what limits the value of W' for the capacitor by considering a small parallel-plate capacitor with plate area A and separation between plates d. For simplicity, we assign the capacitor a voltage rating of 1 V (maximum anticipated voltage across the capacitor). Our goal is to maximize the energy density W'. For a parallel-plate capacitor $C = \varepsilon A/d$, where ε is the permittivity of the insulating material. Using Eq. (4.121) leads to

$$W' = \frac{W}{m} = \frac{1}{2m} CV^2 = \frac{\varepsilon A V^2}{2md} \qquad \text{(J/kg)},$$

where m is the mass of the conducting plates and the insulating material contained in the capacitor. To keep the analysis simple, we assume that the plates can be made so thin as to ignore their masses relative to the mass of the insulating material. If the material's density



Figure TF8-2 Examples of systems that use supercapacitors.

技术简介 8：超级电容器作为电池

作为电子语言的最新补充，超级电容器、超级电容器和纳米电容器等名称表明它们代表了与传统电容器不同或优于传统电容器的设备。这些只是制造商给传统电容器起的奇特名称，还是我们谈论的是一种真正不同类型的电容器？

► 上述三个名称指的是能量存储设备的变体（图TF8-1），其技术名称为电化学双层电容器（EDLC），其中能量存储是通过结合传统静电电容器和电化学伏打电池特征的混合过程实现的。◄

就本技术简介而言，我们将这种相对较新的设备称为超级电容器。电池在储能方面远远优于传统电容器，但电容器的充电和放电速度比电池快得多。作为一种混合技术，超级电容器的功能介于电池和传统电容器之间。超级电容器现在用于支持广泛的应用，从大型发动机（卡车、机车、潜艇等）中的电机启动到数码相机中的闪光灯，并且其使用正在迅速发展

扩展到消费电子产品（手机、MP3播放器、笔记本电脑）和电动汽车（图TF8-2）。

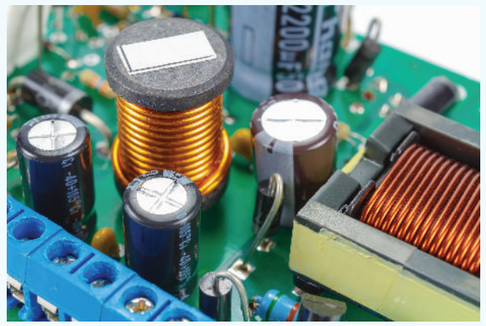


图 TF8-1 超级电容器示例。（由 Vladimir Zhupanenko/Shutterstock 提供。）

电容器储能的局限性

能量密度 W′ 通常以瓦时每千克 (Wh/kg) 为单位进行测量，其中 $1\ \text{Wh} = 3.6 \times 10^3$ 焦耳。因此，设备的能量容量被归一化为其质量。对于电池，W′ 范围从铅酸电池的约 30 Wh/kg 到锂离子电池的高达 150 Wh/kg。相比之下，传统电容器的 W′ 很少超过 0.02 Wh/kg。让我们通过考虑一个具有板面积 A 和板间距 d 的小型平行板电容器来检查限制电容器 W′ 值的因素。为简单起见，我们为电容器指定 1 V 的额定电压（电容器两端的最大预期电压）。我们的目标是最大化能量密度 W′。对于平行板电容器 $C = \varepsilon A/d$，其中 ε 是绝缘材料的介电常数。使用方程式(4.121) 导致

$$W' = \frac{W}{m} = \frac{1}{2m} CV^2 = \frac{\varepsilon A V^2}{2md} \qquad \text{(J/kg)},$$

其中 m 是电容器中包含的导电板和绝缘材料的质量。为了使分析简单，我们假设板可以做得很薄，以忽略它们相对于绝缘材料质量的质量。如果材料的密度



图 TF8-2 使用超级电容器的系统示例。

is ρ (kg/m^3), then $m = \rho A d$ and

$$W' = \frac{\varepsilon V^2}{2\rho d^2} \qquad \text{(J/kg)}.$$

To maximize W', we need to select d to be the smallest possible, but we also have to be aware of the constraint associated with dielectric breakdown. To avoid sparking between the capacitor's two plates, the electric field strength should not exceed $E_{\rm ds}$, which is the dielectric strength of the insulating material. Among the various types of materials commonly used in capacitors, mica has one of the highest values of $E_{\rm ds}$, nearly 2×10^8 V/m. Breakdown voltage $V_{\rm br}$ is related to $E_{\rm ds}$ by $V_{\rm br} = E_{\rm ds} d$, so given that the capacitor is to have a voltage rating of 1 V, let us choose $V_{\rm br}$ to be 2 V, thereby allowing a 50% safety margin. With $V_{\rm br} = 2$ V and $E_{\rm ds} = 2 \times 10^8$ V/m, it follows that d should not be smaller than 10^{-8} m, or 10 nm. For mica, $\varepsilon \approx 6\varepsilon_0$ and $\rho = 3 \times 10^3$ kg/m^3. Ignoring for the moment the practical issues associated with building a capacitor with a spacing of only 10 nm between conductors, the expression for energy density leads to $W' \approx 90$ J/kg. Converting W' to Wh/kg (by dividing by 3.6×10^3 J/Wh) gives

$$W'(\max) = 2.5 \times 10^{-2} \qquad \text{(Wh/kg)},$$

for a traditional capacitor at a voltage rating of 1 V.

► The energy storage capacity of a traditional capacitor is about four orders of magnitude smaller than the energy density capability of a lithium-ion battery. ◄

Energy Storage Comparison

The table in the upper part of **Fig. TF8-3** displays typical values or ranges of values for each of five attributes commonly used to characterize the performance of energy storage devices. In addition to the energy density W', they include the power density P', the charge and discharge rates, and the number of charge/discharge cycles that the device can withstand before deteriorating in performance. For most energy storage devices, the discharge rate usually is shorter than the charge rate, but for the purpose of the present discussion, we treat them as equal. As a first-order approximation, the ***discharge rate*** is related to P' and W' by

$$T = \frac{W'}{P'}.$$

► Supercapacitors are capable of storing 100 to 1000 times more energy than a traditional capacitor but 10 times less than a battery (**Fig. TF8-3**). On the other hand, *supercapacitors can discharge their stored energy in a matter of seconds when compared with hours for a battery.* ◄

Moreover, the supercapacitor's cycle life is on the order of 1 million when compared with only 1000 for a rechargeable battery. Because of these features, the supercapacitor has greatly expanded the scope and use of capacitors in electronic circuits and systems.

Future Developments

The upper right-hand corner of the plot in **Fig. TF8-3** represents the ideal energy storage device with $W' \approx 100$–1000 Wh/kg and $P' \approx 10^3$–10^4 W/kg. The corresponding discharge rate is $T \approx 10$–100 ms. Current research aims to extend the capabilities of batteries and supercapacitors in the direction of this prized domain of the ***energy-power space***.

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是 ρ (kg/m^3)，则 m = ρ Ad 和

$$W' = \frac{\varepsilon V^2}{2\rho d^2} \qquad \text{(J/kg)}.$$

为了最大化 W′，我们需要选择尽可能小的 d，但我们也必须意识到与介电击穿相关的约束。为了避免电容器两块极板之间产生火花，电场强度不应超过 E_{ds}，即绝缘材料的介电强度。在电容器常用的各类材料中，云母是E_{ds}值最高的之一，接近2×10^8V/m。击穿电压 V_{br} 与 E_{ds} 的关系为 $V_{br} = E_{ds}d$，因此考虑到电容器的额定电压为 1 V，我们选择 V_{br} 为 2 V，从而允许 50% 的安全裕度。对于 $V_{br} = 2$ V 和 $E_{ds} = 2 \times 10^8$ V/m，得出 d 不应小于 10^{-8} m 或 10 nm。对于云母，$\varepsilon \approx 6\varepsilon_0$ 和 $\rho = 3 \times 10^3$ kg/m^3。暂时忽略与构建导体间距仅为 10 nm 的电容器相关的实际问题，能量密度的表达式得出 W′ ≈ 90 J/kg。将 W′ 转换为 Wh/kg（除以 3.6×10^3 J/Wh）得出

$$W'(\max) = 2.5 \times 10^{-2} \qquad \text{(Wh/kg)},$$

对于额定电压为 1 V 的传统电容器。

► 传统电容器的能量存储容量比锂离子电池的能量密度小约四个数量级。 ◄

► 超级电容器能够存储比传统电容器多 100 到 1000 倍的能量，但比电池少 10 倍（图 TF8-3）。另一方面，超级电容器可以在几秒钟内释放其存储的能量，而电池则需要数小时。 ◄

此外，超级电容器的循环寿命约为 100 万次，而充电电池的循环寿命仅为 1000 次。由于这些特性，超级电容器极大地扩展了电容器在电子电路和系统中的范围和用途。

未来发展

图 TF8-3 的右上区域代表高比能量、高比功率的理想储能器件：W′ ≈ 100–1000 Wh/kg，P′ ≈ 10^3–10^4 W/kg。研究旨在使电池与超级电容器的性能向该区域发展。

校注： T = W′/P′ 的单位为小时；上述范围对应 0.01–1 h（36–3600 s）。原文“10–100 ms”与所列数值不符。“rate”在此按时间量译为“时间”。

储能比较

图 TF8-3 上方的表格列出储能器件的五类典型性能参数：比能量 W′、比功率 P′、充电时间、放电时间，以及性能衰退前可承受的充放电循环次数。多数器件的放电时间短于充电时间；此处为便于比较，将二者视为相等。在一阶近似下，放电时间 T 与 W′、P′ 的关系为：

$$T = \frac{W'}{P'}.$$

Energy Storage Devices

Feature	Traditional Capacitor	Supercapacitor	Battery
Energy density W' (Wh/kg)	$\sim 10^{-2}$	1 to 10	5 to 150
Power density P' (W/kg)	1,000 to 10,000	1,000 to 5,000	10 to 500
Charge and discharge rate T	10^{-3} sec	$\sim$ 1 sec to 1 min	$\sim$ 1 to 5 hrs
Cycle life N_c	∞	$\sim 10^6$	$\sim 10^3$

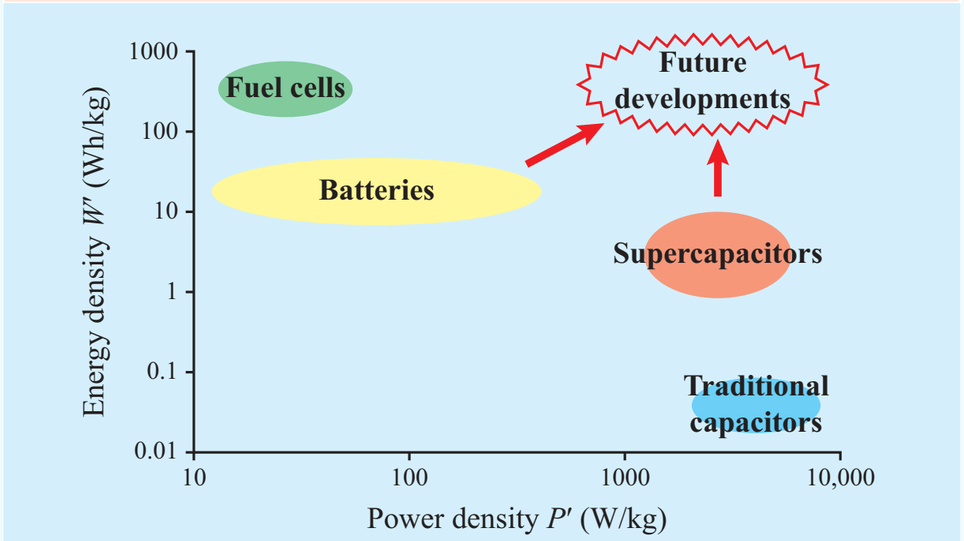
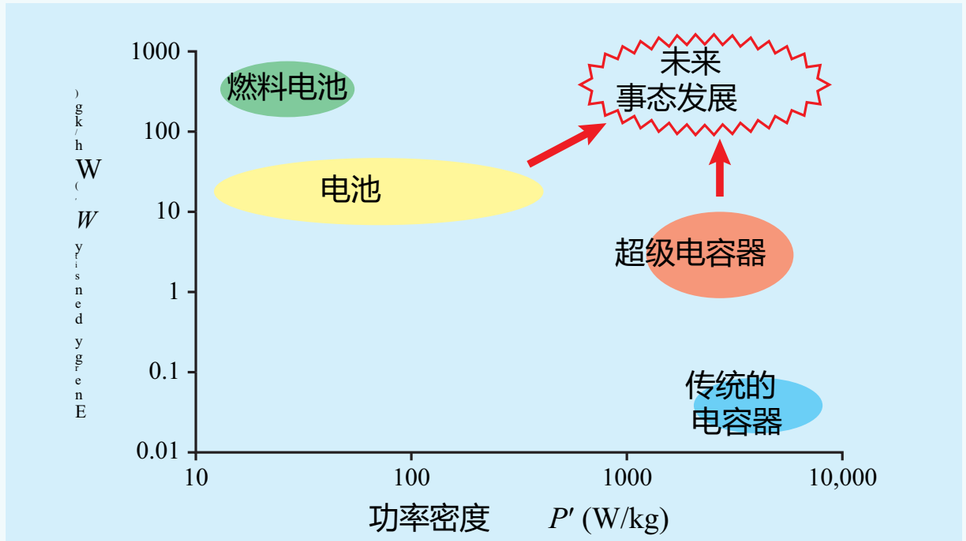


Figure TF8-3 Comparison of energy storage devices.

储能装置

特征		传统电容器	超级电容器	电池
能量密度	W' (Wh/kg)	$\sim 10^{-2}$	1 to 10	5 to 150
功率密度	P' (W/kg)	1,000 to 10,000	1,000 to 5,000	10 to 500
充放电倍率	T	10^{-3} sec	$\sim$ 1 sec to 1 min	$\sim$ 1 to 5 hrs
循环寿命 N_c		∞	$\sim 10^6$	$\sim 10^3$



图TF8-3 储能装置对比。

Table 4-1 Conductivity of some common materials at 20 °C.

Material	Conductivity, σ (S/m)
Conductors	
Silver	6.2×10^7
Copper	5.8×10^7
Gold	4.1×10^7
Aluminum	3.5×10^7
Iron	10^7
Mercury	10^6
Carbon	3×10^4
Semiconductors	
Pure germanium	2.2
Pure silicon	4.4×10^{-4}
Insulators	
Glass	10^{-12}
Paraffin	10^{-15}
Mica	10^{-15}
Fused quartz	10^{-17}

Concept Question 4-14: What are the electromagnetic constitutive parameters of a material?

Concept Question 4-15: What classifies a material as a conductor, a semiconductor, or a dielectric? What is a superconductor?

Concept Question 4-16: What is the conductivity of a perfect dielectric?

4-6.1 Drift Velocity

The drift velocity $\mathbf{u}_e$ of electrons in a conducting material is related to the externally applied electric field $\mathbf{E}$ through

$$\mathbf{u}_e = -\mu_e \mathbf{E} \quad \text{(m/s)}, \tag{4.74a}$$

where μ_e is a material property call the ***electron mobility*** with units of $(m^2/V{\cdot}s)$. In a semiconductor, current flow is due to the movement of both electrons and holes, and since holes are positive-charge carriers, the ***hole drift velocity*** $\mathbf{u}_h$ is in the same direction as $\mathbf{E}$,

$$\mathbf{u}_h = \mu_h \mathbf{E} \quad \text{(m/s)}, \tag{4.74b}$$

where μ_h is the ***hole mobility***. The mobility accounts for the effective mass of a charged particle and the average distance over which the applied electric field can accelerate it before it is stopped by colliding with an atom and then starts accelerating all over again. From Eq. (4.11), the current density in a medium containing a volume density ρ_v of charges moving with velocity $\mathbf{u}$ is $\mathbf{J} = \rho_v \mathbf{u}$. In the most general case, the current density consists of a component $\mathbf{J}_e$ due to electrons and a component $\mathbf{J}_h$ due to holes. Thus, the total ***conduction current density*** is

$$\mathbf{J} = \mathbf{J}_e + \mathbf{J}_h = \rho_{ve}\mathbf{u}_e + \rho_{vh}\mathbf{u}_h \quad \text{(A/m}^2\text{)}, \tag{4.75}$$

where $\rho_{ve} = -N_e e$ and $\rho_{vh} = N_h e$ with N_e and N_h being the number of free electrons and the number of free holes per unit volume and $e = 1.6 \times 10^{-19}$ C is the absolute charge of a single hole or electron. Use of Eqs. (4.74a) and (4.74b) gives

$$\mathbf{J} = (-\rho_{ve}\mu_e + \rho_{vh}\mu_h)\mathbf{E} = \sigma \mathbf{E}, \tag{4.76}$$

where the quantity inside the parentheses is defined as the ***conductivity*** of the material, σ. Thus,

$$\sigma = -\rho_{ve}\mu_e + \rho_{vh}\mu_h = (N_e\mu_e + N_h\mu_h)e \quad \text{(S/m)}, \tag{4.77a}$$
(semiconductor)

and its unit is siemens per meter (S/m). For a good conductor, $N_h\mu_h \ll N_e\mu_e$, and Eq. (4.77a) reduces to

$$\sigma = -\rho_{ve}\mu_e = N_e\mu_e e \quad \text{(S/m)}. \tag{4.77b}$$
(good conductor)

▶ In view of Eq. (4.76), in a ***perfect dielectric*** with $\sigma = 0$, $\mathbf{J} = 0$ regardless of $\mathbf{E}$. Similarly, in a ***perfect conductor*** with $\sigma = \infty$, $\mathbf{E} = \mathbf{J}/\sigma = 0$ regardless of $\mathbf{J}$. ◀

That is,

Perfect dielectric: $\mathbf{J} = 0$,
Perfect conductor: $\mathbf{E} = 0$.

Because σ is on the order of 10^6 S/m for most metals, such as silver, copper, gold, and aluminum (**Table 4-1**), it is common practice to treat them as perfect conductors and to set $\mathbf{E} = 0$ inside them.

A perfect conductor is an ***equipotential*** medium, meaning that the electric potential is the same at every point in the conductor. This property follows from the fact that V_{21}, which is the voltage difference between two points in the conductor,

表 4-1 一些常见材料在 20 °C 时的电导率。

Material	Conductivity, σ (S/m)
Conductors	
Silver	6.2×10^7
Copper	5.8×10^7
Gold	4.1×10^7
Aluminum	3.5×10^7
Iron	10^7
Mercury	10^6
Carbon	3×10^4
Semiconductors	
Pure germanium	2.2
Pure silicon	4.4×10^{-4}
Insulators	
Glass	10^{-12}
Paraffin	10^{-15}
Mica	10^{-15}
Fused quartz	10^{-17}

概念问题 4-14：什么是材料的电磁本构参数？

概念问题 4-15：如何将材料分类为导体、半导体或电介质？什么是超导体？

概念问题 4-16：理想电介质的电导率是多少？

4-6.1 漂移速度

导电材料中电子的漂移速度 u_e 与外部施加的电场 E 相关：

$$\mathbf{u}_e = -\mu_e \mathbf{E} \quad \text{(m/s)}, \tag{4.74a}$$

其中 μ_e 是称为电子迁移率的材料属性，单位为 $(m^2/V{\cdot}s)$。在半导体中，电流是由电子和空穴的运动引起的，并且由于空穴是正电荷载流子，因此空穴漂移速度 u_h 与 E 方向相同，

$$\mathbf{u}_h = \mu_h \mathbf{E} \quad \text{(m/s)}, \tag{4.74b}$$

其中 μ_h 是空穴迁移率。迁移率考虑了带电粒子的有效质量和平均距离

在它与原子碰撞而停止之前，所施加的电场可以对其进行加速，然后再次开始加速。从方程。(4.11) 中，含有以速度 u 运动的体积密度 ρ_v 的电荷的介质中的电流密度为 $J = \rho_v u$。在最一般的情况下，电流密度由电子引起的分量 J_e 和空穴引起的分量 J_h 组成。因此，总传导电流密度为

$$\mathbf{J} = \mathbf{J}_e + \mathbf{J}_h = \rho_{ve}\mathbf{u}_e + \rho_{vh}\mathbf{u}_h \quad \text{(上午}^2\text{)}, \tag{4.75}$$

其中 $\rho_{ve} = -N_e e$ 和 $\rho_{vh} = N_h e$，其中 N_e 和 N_h 是每单位体积的自由电子数量和自由空穴数量，$e = 1.6 \times 10^{-19}$ C 是单个空穴或电子的绝对电荷。使用等式。(4.74a) 和 (4.74b) 给出

$$\mathbf{J} = (-\rho_{ve}\mu_e + \rho_{vh}\mu_h)\mathbf{E} = \sigma \mathbf{E}, \tag{4.76}$$

其中括号内的数量定义为材料的电导率，σ。因此，

$$\sigma = -\rho_{ve}\mu_e + \rho_{vh}\mu_h = (N_e\mu_e + N_h\mu_h)e \quad \text{(S/m)}, \tag{4.77a}$$
(semiconductor)

单位为西门子每米（S/m）。对于良导体，$N_h\mu_h \ll N_e\mu_e$ 和方程：(4.77a) 减少为

$$\sigma = -\rho_{ve}\mu_e = N_e\mu_e e \quad \text{(S/m)}. \tag{4.77b}$$
(good conductor)

▶ 鉴于方程。（4.76），在具有 $\sigma = 0$ 的理想电介质中，无论 E 为何，$J = 0$。类似地，在具有 $\sigma = \infty$ 的理想导体中，无论 J 为何，$E = J/\sigma = 0$。◀

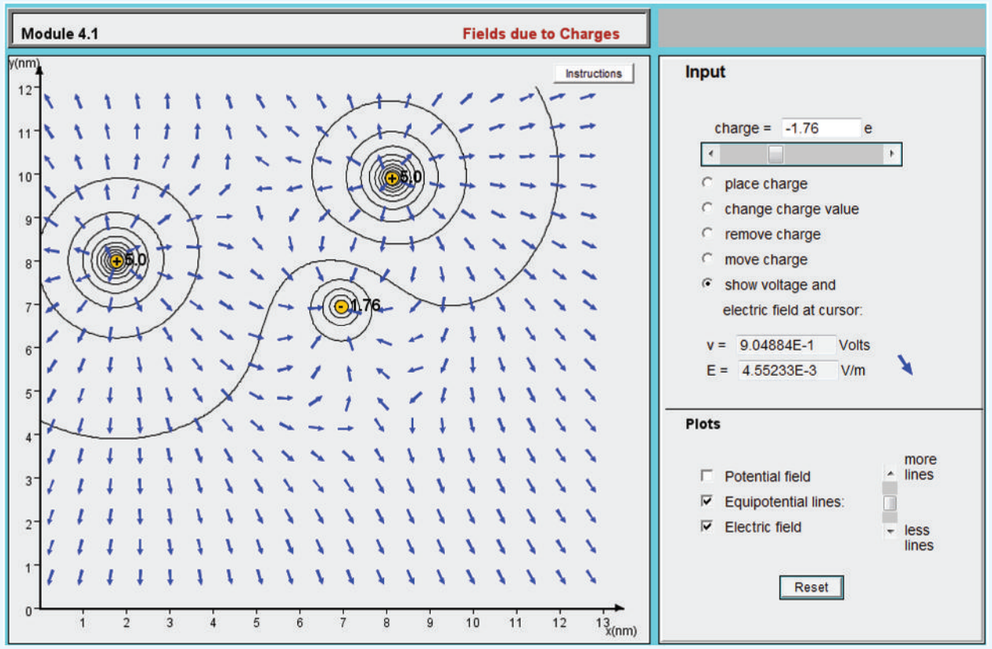
那是，

Perfect dielectric: $\mathbf{J} = 0$,
Perfect conductor: $\mathbf{E} = 0$.

由于大多数金属（例如银、铜、金和铝）的 σ 约为 10^6 S/m（表 4-1），因此通常的做法是将它们视为理想导体并在其中设置 $E = 0$。

理想导体是等电位介质，这意味着导体中每一点的电位都相同。此属性源自以下事实：V_{21}，即导体中两点之间的电压差，

Module 4.1 Fields Due to Charges For any group of point charges, this module calculates and displays the electric field **E** and potential V across a 2-D grid. The user can specify the locations, magnitudes and polarities of the charges.



equals the line integral of **E** between them, as indicated by Eq. (4.39). Since $\mathbf{E} = 0$ everywhere in the perfect conductor, the voltage difference $V_{21} = 0$. The fact that the conductor is an equipotential medium, however, does not necessarily imply that the potential difference between the conductor and some other conductor is zero. Each conductor is an equipotential medium, but the presence of different distributions of charges on their two surfaces can generate a potential difference between them.

Example 4-10: Conduction Current in a Copper Wire

A 2-mm diameter copper wire with conductivity of 5.8×10^7 S/m and electron mobility of 0.0032 (m^2/V·s) is subjected to an electric field of 20 (mV/m). Find (a) the volume charge density of the free electrons, (b) the current density, (c) the current flowing in the wire, (d) the electron drift velocity, and (e) the volume density of the free electrons.

Solution: (a)

$$\rho_{ve} = -\frac{\sigma}{\mu_e} = -\frac{5.8 \times 10^7}{0.0032} = -1.81 \times 10^{10}\ (\text{C/m}^3).$$

(b)

$$J = \sigma E = 5.8 \times 10^7 \times 20 \times 10^{-3} = 1.16 \times 10^6\ (\text{A/m}^2).$$

(c)

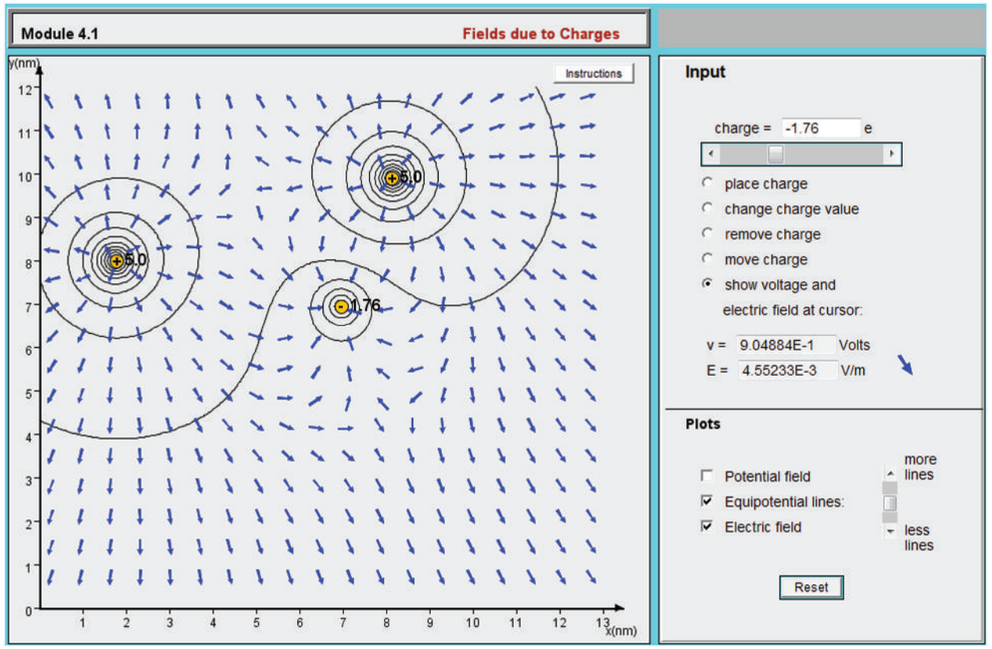
$$I = JA$$
$$= J\left(\frac{\pi d^2}{4}\right) = 1.16 \times 10^6 \left(\frac{\pi \times 4 \times 10^{-6}}{4}\right) = 3.64\ \text{A}.$$

(d)

$$u_e = -\mu_e E = -0.0032 \times 20 \times 10^{-3} = -6.4 \times 10^{-5}\ \text{m/s}.$$

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模块 4.1 电荷产生的场 对于任何一组点电荷，该模块计算并显示二维网格上的电场 E 和电势 V。用户可以指定电荷的位置、大小和极性。



等于它们之间 E 的线积分，如式（1）所示。 （4.39）。由于理想导体中到处 E = 0，因此电压差 $V_{21} = 0$。然而，导体是等电位介质的事实并不一定意味着该导体与其他导体之间的电位差为零。每个导体都是等电位介质，但其两个表面上不同电荷分布的存在会在它们之间产生电位差。

例 4-10：铜线中的传导电流

一根直径为 2 毫米的铜线，其电导率为 5.8×10^7 S/m，电子迁移率为 0.0032 (m^2/V·s)，该铜线承受 20 (mV/m) 的电场。求 (a) 自由电子的体积电荷密度，(b) 电流密度，(c) 导线中流动的电流，(d) 电子漂移速度，以及 (e) 自由电子的体积密度。

解：（一）

$$\rho_{ve} = -\frac{\sigma}{\mu_e} = -\frac{5.8 \times 10^7}{0.0032} = -1.81 \times 10^{10}\ (\text{C/m}^3).$$

(b)

$$J = \sigma E = 5.8 \times 10^7 \times 20 \times 10^{-3} = 1.16 \times 10^6\ (\text{A/m}^2).$$

(c)

$$I = JA$$
$$= J\left(\frac{\pi d^2}{4}\right) = 1.16 \times 10^6 \left(\frac{\pi \times 4 \times 10^{-6}}{4}\right) = 3.64\ \text{A}.$$

(d)

$$u_e = -\mu_e E = -0.0032 \times 20 \times 10^{-3} = -6.4 \times 10^{-5}\ \text{m/s}.$$